THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

TEAM AWARD HOMEWORK SET #7

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2017
Professor: Dr. Michael Caracotsios
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Your name: Megan Young
Your name: Zanub Hassan
Your name: Reem Khoury
Due Date: Wednesday Nov-22, 2017 in class

Number of points.................................100
Your score...............................

Award: Gift card to each member of the winning team

To be considered for the award, the team homework final report must be typed. Homework results should be presented in a logical and concise manner, with all steps in derivation processes [if any] shown very clearly. The ASPEN PFD with critical data must be printed in a full page and included with the homework solution report. Images of the Block Results as well as the Stream Results must also be clearly shown in the homework solution report. You may also want to include anything else (like external references) you believe will increase your chances of winning the award.

Remember: All members of the team that wins a total of three(3) awards will be given **100** points towards the final examination and will be excused from taking it.

Problem 1(100 pts): Aspen Plus Simulation of CO_2 Removal from Syngas/Natural Gas

Honeywell UOP

PETRONAS' floating liquefied natural gas (FLNG) facility, the first of its kind in the world, uses Honeywell UOP's Amine Guard FS™ technology, specially adapted for marine applications

World's first floating LNG facility-PETRONAS

Syngas is usually produced from gasification of coal or steam methane reforming. Depending on the source of production it contains various amounts for carbon dioxide that needs to be removed prior to further purification for production of hydrogen. The removal of CO2 is usually accomplished with chemical solvents, such as amines. The most common amine solvent is MDEA, methyl di-ethanolamine. A typical process for the removal of CO2 is shown in the process flowsheet diagram below.

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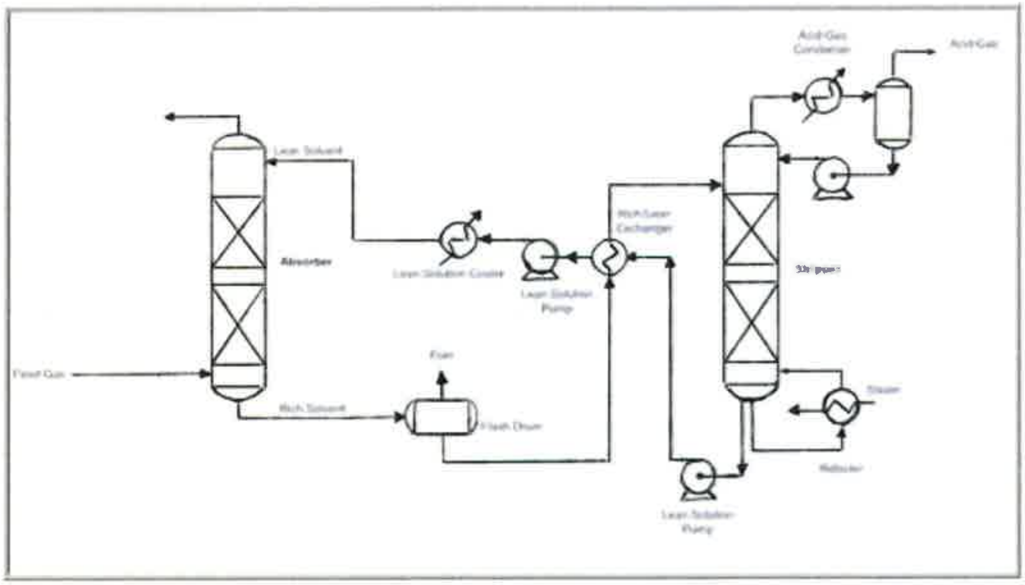
HOMEWORK STATEMENTS:



The following data are available for this process.

Syngas Feed	*Value*	Absorber	*Value*	Stripper	*Value*
Temperature, °C	40.0	Number of Stages	15	Number of Stages	18
Pressure, bar	55.0	Top Pressure, bar	53.0	Column Pressure drop, bar	0.3
Flow, kmol/hr	2,000.0	Pressure drop			
CO, mole %	29.1	HETP, m	1.2	**Condenser**	
CO_2	10.0	Diameter, m		Pressure, bar	2.0
Hydrogen	59.8	**Packing**		Temperature, °C	
H_2O	0.20	Raschig Metal			
Nitrogen	0.30	Dimension, mm	35.0	**Pumps**	
Argon	0.10			LP Lean Solution, bar	15.0
Methane	0.50			HP Lean Solution, bar	55.0

The Rich Solvent flash drum should operate at a pressure in the range of 10-20 bar. You may wish to do some research to find out what would be the optimal pressure. Make sure you have enough pressure after the drum to bring the rich solution to the feed point at the top of the stripper. To estimate the discharge pressure of the reflux pump, you will have to calculate an approximate height of the stripper column and assume that the reflux pump is at zero elevation. For the Rich/Lean Exchanger, the stripper column condenser and the Lean Solvent cooler you may assume a $\Delta T_{min} = 10^{o}C$. Pressure drop for the Rich/Lean heat exchanger is about 5 psi on the shell side and 10 psi on the tube side. Make sure you make the proper selection of fluid for the shell and tube sides. Cooling water is available at $95^{o}F$. If you need

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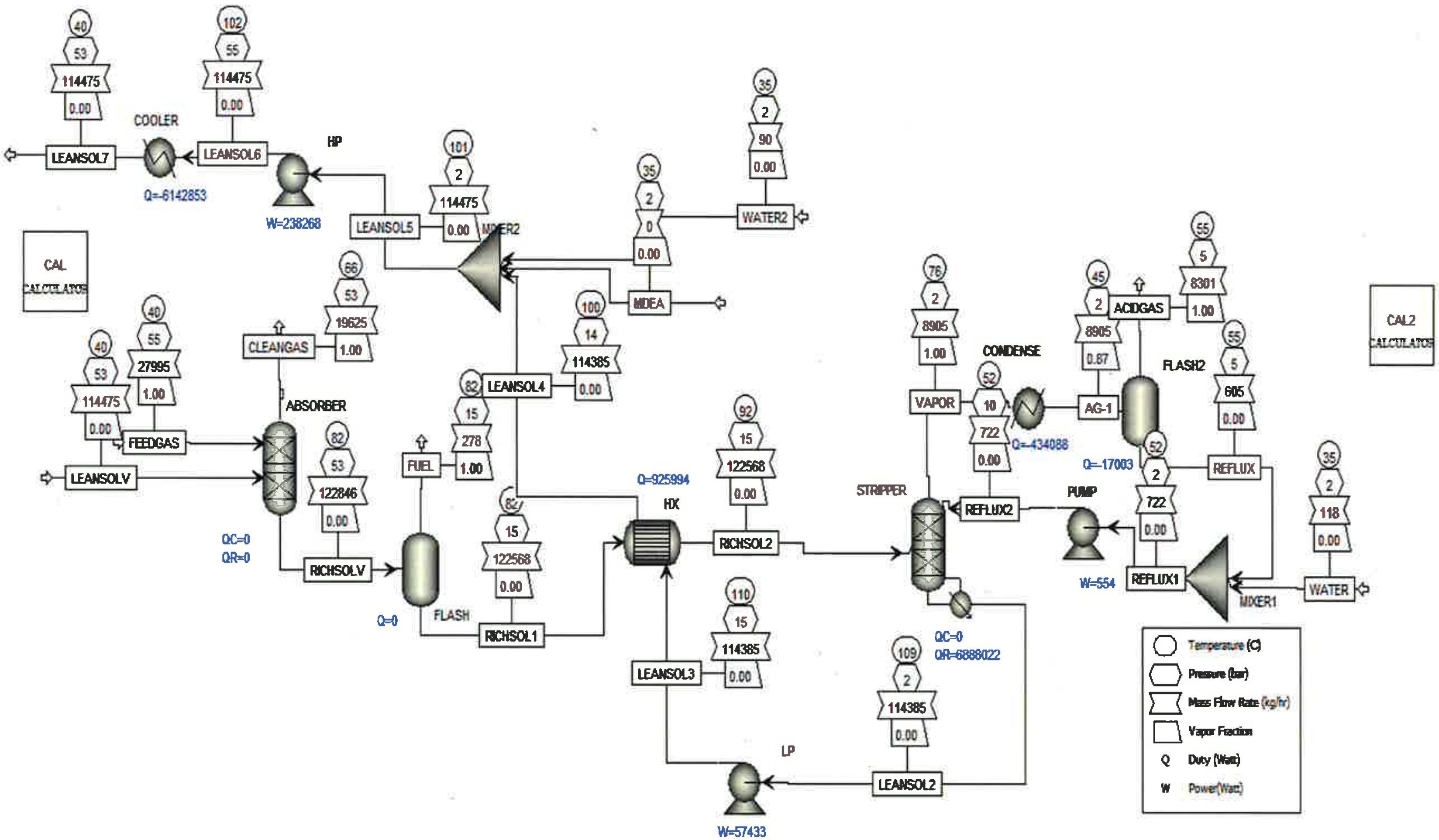
other data, you will need to do some research on your own. Make sure everyone in the team does his/her own part.

OBJECTIVES OF THIS HOMEWORK

1. Prepare an ASPEN Plus Simulation for this process. You may start with the basic template that I have created for you and contains the physical properties and the amine chemistry
2. Your objective is to calculate the optimal amounts of the lean solvent and the reboiler duty in order to make the operating cost reasonable and to obtain a molar composition of carbon dioxide in the clean gas of 1%. Assume that cost of cooling is **0.2$/GJ**, the cost of heating is **$5.5/GJ**, and cost of electricity is **$0.06/kWh**.
3. The lean solvent has a molar water to amine ratio of 6.615. The load is something that you have to decide
4. The team that arrives at the smallest cost and meets the specifications of the clean gas will get the award.

PROBLEM 1:



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Data Input:

COMPONENTS
CARBON MONOXIDE
CARBON DIOXIDE
HYDROGEN
WATER
NITROGEN
ARGON
METHANE
MDEA
MDEA+
H3O+
OH-
HCO3-
CO3--

PROPERTY METHODS	
METHOD FILTER	Common
METHOD NAME	ELECTNRTL
BASE METHOD	ELECTNRTL
FREE WATER METHOD	Steam-TA
WATER SOLUBILITY	3

STREAM INPUT	FEEDGAS
TEMPERATURE (°C)	40
PRESSURE (BAR)	55
FLOW RATE (KMOL/H)	2,000
MOLAR COMPOSITION %	
CARBON MONOXIDE	29.1
CARBON DIOXIDE	10.0
HYDROGEN	59.8
WATER	0.20
NITROGEN	0.30
ARGON	0.10
METHANE	0.50

STREAM INPUT	LEANSOLV
TEMPERATURE (°C)	40
PRESSURE (BAR)	53
FLOW RATE (KMOL/H)	800 (Guessed)
MOLE-FLOW (KMOL/H)	
CARBON DIOXIDE	8 (Guessed)
WATER	688 (Guessed)
MDEA	104 (Guessed)

STREAM INPUT	WATER
TEMPERATURE (°C)	35
PRESSURE (BAR)	2
FLOW RATE (KMOL/H)	100 (Guessed)
MOLE-FRAC	
WATER	1.0

STREAM INPUT	WATER2
TEMPERATURE (°C)	35
PRESSURE (BAR)	2
FLOW RATE (KMOL/H)	1000 (Guessed)
MOLE-FRAC	
WATER	1.0

STREAM INPUT	MDEA
TEMPERATURE (°C)	35
PRESSURE (BAR)	2
FLOW RATE (KMOL/H)	1000 (Guessed)
MOLE-FRAC	
MDEA	1.0

Block Inputs:

BLOCK INPUT – RADFRAC	ABSORBER	
CALCULATION TYPE	Equilibrium	
NUMBER OF STAGES	15	
CONDENSER	None	
REBOILER	None	
VALID PHASES	Vapor-Liquid	
CONVERGENCE	Standard	
OPERATING SPECS		
FREE WATER REFLUX RATIO	0	
FEED STREAMS	STAGE	CONVENTION
FEEDGAS	16	Above-Stage
LEANSOLV	1	Above-Stage
PRODUCT STREAMS	STAGE	PHASE
RICHSOLV	15	Liquid
CLEANGAS	1	Vapor
PRESSURE		
STAGE 1/CONDENSOR PRESSURE	53	bar

BLOCK INPUT – FLASH2	FLASH
FLASH TYPE	
PRESSURE (BAR)	15
DUTY (WATT)	0
VALID PHASES	Vapor-Liquid

BLOCK INPUT – HEATEXCHANGER	HX
SPECIFICATIONS	
COLD STREAM TEMPERATURE INCREASE (°C)	10
PRESSURE DROP	
HOT SIDE (TUBE) (BAR)	14.311
COLD SIDE (SHELL) (BAR)	14.655

BLOCK INPUT – HEATER	CONDENSE
FLASH SPECS	
TEMPERATURE (°C)	45
PRESSURE (BAR)	2
VALID PHASES	Vapor-Liquid

BLOCK INPUT – FLASH2	FLASH2
FLASH SPECS	
TEMPERATURE (°C)	55
PRESSURE (BAR)	5

BLOCK INPUT – PUMP	PUMP
DISCHARGE PRESSURE (BAR)	10

BLOCK INPUT – PUMP	LP
DISCHARGE PRESSURE (BAR)	15

BLOCK INPUT – PUMP	HP
DISCHARGE PRESSURE (BAR)	55

BLOCK INPUT - RADFRAC	STRIPPER	
CALCULATION TYPE	Equilibrium	
NUMBER OF STAGES	18	
CONDENSER	None	
REBOILER	Kettle	
VALID PHASES	Vapor-Liquid	
CONVERGENCE	Standard	
OPERATING SPECS		
REBOILER DUTY (BTU/HR)	1.2e+07 (Guessed)	
FREE WATER REFLUX RATIO	0	
FEED STREAMS	**STAGE**	**CONVENTION**
REFLUX2	1	Above-Stage
RICHSOL2	1	Above-Stage
PRODUCT STREAMS	**STAGE**	**PHASE**
LEANSOL2	18	Liquid
VAPOR	1	Vapor
PRESSURE		
STAGE 1/CONDENSOR PRESSURE	2	bar
COLUMN PRESSURE DROP	0.3	bar

BLOCK INPUT – HEATER	COOLER
FLASH SPECS	
TEMPERATURE (°C)	40
PRESSURE (BAR)	53
VALID PHASES	Vapor-Liquid

Calculator Block Inputs:

Calculator blocks were used to calculate the composition of the lean solvent stream (LEANSOLV) when the total molar flowrate was changed. The load (CO_2 / MDEA ratio) was chosen to be 0.08. Therefore, to calculate the molar flowrates of each component, we use the following equations:

$$\frac{Water\ Molar\ Flowrate}{Amine\ Molar\ Flowrate} = 6.615 \rightarrow WATER = 6.615 * MDEA$$

$$\frac{CO_2\ Molar\ Flowrate}{MDEA\ Molar\ Flowrate} = 0.08 \rightarrow CO_2 = 0.08 * MDEA$$

$$TOTAL\ FLOWRATE = MDEA + WATER + CO_2 = MDEA + 6.615 * MDEA + 0.08 * MDEA$$

$$MDEA = \frac{TOTAL}{7.695}$$

These equations were entered into the fortran tab of the calculator block. The molar flowrate compositions of the lean solvent are then calculated each time the total molar flowrate is changed and also exported into the stream input.

CALCULATOR BLOCK INPUT					**CAL**		
DEFINE VARIABLES	INFO FLOW	CATEGORY	TYPE	STREAM	SUBSTREAM	VARIABLE/COMPONENT	UNITS
TOTAL	IMPORT	Streams	Stream-Var	LEANSOLV	MIXED	MOLE-FLOW	kmol/hr
MDEA	EXPORT	Streams	Mole-Flow	LEANSOLV	MIXED	C5H13NO2	kmol/hr
WATER	EXPORT	Streams	Mole-Flow	LEANSOLV	MIXED	H2O	kmol/hr
CO2	EXPORT	Streams	Mole-Flow	LEANSOLV	MIXED	CO2	kmol/hr
CALCULATE							
CALCULATE METHOD	Fortan						
	MDEA = TOTAL / 7,695						
	WATER = 6.615 * MDEA						
	CO2 = 0.08 * MDEA						
SEQUENCE							
EXECUTE	Use Import/Export Variables						
IMPORT VARIABLES	TOTAL						
EXPORT VARIABLES	MDEA	WATER	CO2				

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The second calculator block (CAL2) was used for the water makeup stream into the reflux of the stripper column. Water is lost when the vapor outlet stream from the stripper is flashed, removing acid gas. The water makeup is calculated by subtracting the water molar flowrate of the vapor outlet (AG-1) from the reflux stream (REFLUX). The makeup flowrate is exported into the WATER stream.

CALCULATOR BLOCK INPUT					**CAL2**		
DEFINE VARIABLES	INFO FLOW	CATEGORY	TYPE	STREAM	SUBSTREAM	COMPONENT	UNITS
W1	IMPORT	Streams	Mole-Flow	AG-1	MIXED	H2O	kmol/hr
W2	IMPORT	Streams	Mole-Flow	REFLUX	MIXED	H2O	kmol/hr
W3	EXPORT	Streams	Mole-Flow	WATER	MIXED	H2O	kmol/hr
CALCULATE							
CALCULATE METHOD	Fortan						
	W3 = W1 – W2						
SEQUENCE							
EXECUTE	Use Import/Export Variables						
IMPORT VARIABLES	W1	W2					
EXPORT VARIABLES	W3						

Design Spec Inputs:

Design Spec 1 (DS-1) was used to find the molar flowrate of the lean solvent (LEANSOLV) in order to have 1 mol% of CO_2 in the clean gas stream out of the absorber (CLEANGAS). DS-2 was used to find the reboiler duty of the stripper that will give the same CO_2 composition of the lean solvent into the absorber (LEANSOLV) and the lean solvent out of the stripper (LEANSOL2). DS-3 was used to find the molar flowrate of water makeup before entering the absorber column (LEANSOL2), which needed to be the same composition as LEANSOLV. DS-4 was also used to make the LEANSOL2 the same composition as LEANSOLV by adding makeup MDEA.

DESIGN SPEC INPUT			**DS-1**		
DEFINE VARIABLES	CATEGORY	TYPE	STREAM	SUBSTREAM	COMPONENT
C1	Streams	Mole-Frac	CLEANGAS	MIXED	CO2
CALCULATE					
SPEC	C1	=	0.01		
TOLERANCE	0.001				
VARY					
TYPE	STREAM	SUBSTREAM	VARIABLE	UNITS	
STREAM-VAR	LEANSOLV	MIXED	MOLE-FLOW	kmol/hr	
LOWER LIMIT	1500				
UPPER LIMIT	4000				

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DESIGN SPEC INPUT			**DS-2**			
DEFINE VARIABLES	CATEGORY	TYPE	STREAM	SUBSTREAM	COMPONENT	UNITS
C2	Streams	Mole-Flow	LEANSOLV	MIXED	CO2	kmol/hr
C3	Streams	Mole-Flow	LEANSOL2	MIXED	CO2	kmol/hr
CALCULATE						
SPEC	C2	=	C3			
TOLERANCE	0.01					
VARY						
TYPE	BLOCK	VARIABLE	UNITS			
BLOCK-VAR	STRIPPER	QN	Watts			
LOWER LIMIT	1					
UPPER LIMIT	2e7					

DESIGN SPEC INPUT			**DS-3**			
DEFINE VARIABLES	CATEGORY	TYPE	STREAM	SUBSTREAM	COMPONENT	UNITS
W1	Streams	Mole-Flow	LEANSOLV	MIXED	H2O	kmol/hr
W2	Streams	Mole-Flow	LEANSOL2	MIXED	H2O	kmol/hr
CALCULATE						
SPEC	W1	=	W1			
TOLERANCE	0.001					
VARY						
TYPE	STREAM	SUBSTREAM	VARIABLE	UNITS		
STREAM-VAR	WATER2	MIXED	MOLE-FLOW	kmol/hr		
LOWER LIMIT	1					
UPPER LIMIT	1000					

DESIGN SPEC INPUT				DS-4		
DEFINE VARIABLES	CATEGORY	TYPE	STREAM	SUBSTREAM	COMPONENT	UNITS
M1	Streams	Mole-Flow	LEANSOLV	MIXED	C5H13NO2	kmol/hr
M2	Streams	Mole-Flow	LEANSOL2	MIXED	C5H13NO2	kmol/hr
CALCULATE						
SPEC	M1	=	M2			
TOLERANCE	0.001					
VARY						
TYPE	STREAM	SUBSTREAM	VARIABLE	UNITS		
STREAM-VAR	MDEA	MIXED	MOLE-FLOW	kmol/hr		
LOWER LIMIT	0					
UPPER LIMIT	1					

Stream Results:

	ACIDGAS	AG-1	CLEANGAS	FEEDGAS	FUEL	LEANSOL2	LEANSOL3
TEMPERATURE C	55	45	65.5	40	81.5	109.3	109.7
PRESSURE BAR	5	2	53	55	15	2.3	15
VAPOR FRAC	1	0.87	1	1	1		
MOLE FLOW KMOL/HR	198.285	231.732	1803.884	2000	9.392	3637.184	3637.184
MOLE FLOW TOTAL MMSCFD	3.981	4.653	36.22	40.158	0.189	73.03	73.03
MASS FLOW KG/HR	8300.608	8905.267	19625.349	27995.47	278.008	114384.647	114384.647
VOLUME FLOW CUM/HR	1062.455	2646.403	980.691	960.528	18.27	116.476	116.483
ENTHALPY GCAL/HR	-17.397	-19.649	-16.92	-34.437	-0.399	-262.045	-261.996
CO (MASS FLOW KG/HR)	306.597	306.86	15837.266	16302.05	158.189		
CO_2	7826.309	7828.663	865.923	8801.96	109.823	1666.248	1666.248
H_2	0.07	0.07	2409.309	2410.992	1.613		
H_2O	117.731	718.532	157.407	72.061	5.131	56315.82	56315.82
N_2	0.002	0.002	168.009	168.081	0.07		
AR	4.603	4.613	74.311	79.896	0.982		
CH_4	45.296	45.732	112.938	160.428	2.194		
$C_5H_{13}NO_2$	0	0.795	0.184	0	0.007	56402.579	56402.579
CO (MASS FRAC)	0.037	0.034	0.807	0.582	0.569		
CO_2	0.943	0.879	0.044	0.314	0.395	0.015	0.015
H_2	0	0	0.123	0.086	0.006		
H_2O	0.014	0.081	0.008	0.003	0.018	0.492	0.492
N_2	0	0	0.009	0.006	0		
AR	0.001	0.001	0.004	0.003	0.004		
CH_4	0.005	0.005	0.006	0.006	0.008		
$C_5H_{13}NO_2$						0.493	0.493
CO (MASS FLOW KMOL/HR)	10.946	10.955	565.407	582	5.648		
CO_2	177.831	177.885	19.676	200	2.495	37.861	37.861
H_2	0.035	0.035	1195.165	1196	0.8		
H_2O	6.535	39.885	8.737	4	0.285	3126.003	3126.003
N_2			5.997	6	0.002		
AR	0.115	0.115	1.86	2	0.025		
CH_4	2.823	2.851	7.04	10	0.137		
$C_5H_{13}NO_2$		0.007	0.002			473.32	473.32
CO (MOLEFRAC)	0.055	0.047	0.313	0.291	0.601		
CO_2	0.897	0.768	0.011	0.1	0.266	0.01	0.01
H_2			0.663	0.598	0.085		
H_2O	0.033	0.172	0.005	0.002	0.03	0.859	0.859
N_2			0.003	0.003			
AR	0.001		0.001	0.001	0.003		
CH_4	0.014	0.012	0.004	0.005	0.015		
$C_5H_{13}NO_2$						0.13	0.13

	LEANSOL4	LEANSOL5	LEANSOL6	LEANSOL7	LEANSOLV	MDEA
TEMPERATURE C	100.4	100.5	102.3	40	40	35
PRESSURE BAR	14.311	2	55	53	53	2
VAPOR FRAC						
MOLE FLOW KMOL/HR	3637.184	3642.204	3642.204	3642.204	3642.209	0.002
MOLE FLOW TOTAL MMSCFD	73.03	73.131	73.131	73.131	73.131	0
MASS FLOW KG/HR	114384.647	114475.25	114475.25	114475.251	114475.466	0.191
VOLUME FLOW CUM/HR	115.554	115.691	115.712	110.46	110.46	
ENTHALPY GCAL/HR	-262.792	-263.134	-262.929	-268.211	-268.212	
CO (MASS FLOW KG/HR)						
CO2	1666.248	1666.248	1666.248	1666.248	1666.463	
H2						
H2O	56315.82	56406.233	56406.233	56406.233	56406.233	
N2						
AR						
CH4						
C5H13NO2	56402.579	56402.77	56402.77	56402.77	56402.77	0.191
CO (MASS FRAC)						
CO2	0.015	0.015	0.015	0.015	0.015	
H2						
H2O	0.492	0.493	0.493	0.493	0.493	
N2						
AR						
CH4						
C5H13NO2	0.493	0.493	0.493	0.493	0.493	1
CO (MASS FLOW KMOL/HR)						
CO2	37.861	37.861	37.861	37.861	37.866	
H2						
H2O	3126.003	3131.022	3131.022	3131.022	3131.022	
N2						
AR						
CH4						
C5H13NO2	473.32	473.322	473.322	473.322	473.322	0.002
CO (MOLEFRAC)						
CO2	0.01	0.01	0.01	0.01	0.01	
H2						
H2O	0.859	0.86	0.86	0.86	0.86	
N2						
AR						
CH4						
C5H13NO2	0.13	0.13	0.13	0.13	0.13	1

	REFLUX	REFLUX1	REFLUX2	RICHSOL1	RICHSOL2
TEMPERATURE C	55	51.6	52.3	81.5	91.5
PRESSURE BAR	5	2	10	15	14.655
VAPOR FRAC		0.001			0.003
MOLE FLOW KMOL/HR	33.447	39.982	39.982	3828.933	3828.933
MOLE FLOW TOTAL MMSCFD	0.672	0.803	0.803	76.88	76.88
MASS FLOW KG/HR	604.659	722.4	722.4	122567.58	122567.58
VOLUME FLOW CUM/HR	0.618	1.136	0.736	112.438	139.015
ENTHALPY GCAL/HR	-2.267	-2.712	-2.712	-285.329	-284.533
CO (MASS FLOW KG/HR)	0.263	0.263	0.263	306.597	306.597
CO2	2.354	2.354	2.354	9492.677	9492.677
H2				0.07	0.07
H2O	600.801	718.542	718.542	56315.756	56315.756
N2				0.002	0.002
AR	0.009	0.009	0.009	4.603	4.603
CH4	0.436	0.436	0.436	45.296	45.296
C5H13NO2	0.795	0.795	0.795	56402.579	56402.579
CO (MASS FRAC)				0.003	0.003
CO2	0.004	0.003	0.003	0.077	0.077
H2					
H2O	0.994	0.995	0.995	0.459	0.459
N2					
AR					
CH4	0.001	0.001	0.001		
C5H13NO2	0.001	0.001	0.001	0.46	0.46
CO (MASS FLOW KMOL/HR)	0.009	0.009	0.009	10.946	10.946
CO2	0.053	0.053	0.053	215.695	215.695
H2	0	0	0	0.035	0.035
H2O	33.35	39.885	39.885	3125.999	3125.999
N2					
AR				0.115	0.115
CH4	0.027	0.027	0.027	2.823	2.823
C5H13NO2	0.007	0.007	0.007	473.32	473.32
CO (MOLEFRAC)				0.003	0.003
CO2	0.002	0.001	0.001	0.056	0.056
H2					
H2O	0.997	0.998	0.998	0.816	0.816
N2					
AR					
CH4	0.001	0.001	0.001	0.001	0.001
C5H13NO2				0.124	0.124

	RICHSOLV	VAPOR	WATER	WATER2
TEMPERATURE C	82	75.9	35	35
PRESSURE BAR	53	2	2	2
VAPOR FRAC		1		
MOLE FLOW KMOL/HR	3838.325	231.732	6.535	5.019
MOLE FLOW TOTAL MMSCFD	77.069	4.653	0.131	0.101
MASS FLOW KG/HR	122845.588	8905.267	117.731	90.413
VOLUME FLOW CUM/HR	112.832	3337.713	0.118	0.091
ENTHALPY GCAL/HR	-285.728	-19.276	-0.445	-0.342
CO (MASS FLOW KG/HR)	464.786	306.86		
CO2	9602.5	7828.663		
H2	1.683	0.07		
H2O	56320.887	718.532	117.731	90.413
N2	0.072	0.002		
AR	5.585	4.613		
CH4	47.49	45.732		
C5H13NO2	56402.586	0.795		
CO (MASS FRAC)	0.004	0.034		
CO2	0.078	0.879		
H2				
H2O	0.458	0.081		
N2				
AR		0.001		
CH4		0.005		
C5H13NO2	0.459			
CO (MASS FLOW KMOL/HR)	16.593	10.955		
CO2	218.19	177.885		
H2	0.835	0.035		
H2O	3126.284	39.885	6.535	5.019
N2	0.003	0		
AR	0.14	0.115		
CH4	2.96	2.851		
C5H13NO2	473.32	0.007		
CO (MOLEFRAC)	0.004	0.047		
CO2	0.057	0.768		
H2				
H2O	0.814	0.172	1	1
N2				
AR				
CH4	0.001	0.012		
C5H13NO2	0.123			

Block Results:

CONDENSER & COOLER

NAME	CONDENSE	COOLER
PROPERTY METHOD	ELECNRTL	ELECNRTL
HENRY'S COMPONENT LIST ID	GLOBAL	GLOBAL
ELECTROLYTE CHEMISTRY ID	GLOBAL	GLOBAL
USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO	NO
FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA	STEAM-TA
WATER SOLUBILITY METHOD	3	3
SPECIFIED PRESSURE [BAR]	2	53
SPECIFIED TEMPERATURE [C]	45	40
CALCULATED PRESSURE [BAR]	2	53
CALCULATED TEMPERATURE [C]	45	40
CALCULATED VAPOR FRACTION	0.87028282	0
CALCULATED HEAT DUTY [WATT]	-434088.151	-6142853.21
TEMPERATURE CHANGE [C]	-10	
NET DUTY [WATT]	-434088.151	-6142853.21
FIRST LIQUID / TOTAL LIQUID	1	1
TOTAL FEED STREAM CO2E FLOW [KG/HR]	8971.96262	666.24798
TOTAL PRODUCT STREAM CO2E FLOW [KG/HR]	8971.96262	1666.24798

HEAT EXCHANGER

NAME	HX
HOT SIDE PROPERTY METHOD	ELECNRTL
HOT SIDE HENRY'S COMPONENT LIST ID	GLOBAL
HOT SIDE ELECTROLYTE CHEMISTRY ID	GLOBAL
HOT SIDE USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO
HOT SIDE FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA
HOT SIDE WATER SOLUBILITY METHOD	3
COLD SIDE PROPERTY METHOD	ELECNRTL
COLD SIDE HENRY'S COMPONENT LIST ID	GLOBAL
COLD SIDE ELECTROLYTE CHEMISTRY ID	GLOBAL
COLD SIDE USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO
COLD SIDE FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA
COLD SIDE WATER SOLUBILITY METHOD	3
EXCHANGER SPECIFICATION	10
MINIMUM TEMPERATURE APPROACH [C]	1
HOT SIDE OUTLET PRESSURE [BAR]	14.311
COLD SIDE OUTLET PRESSURE [BAR]	14.655
INLET HOT STREAM TEMPERATURE [C]	109.747688
INLET HOT STREAM PRESSURE [BAR]	15
OUTLET HOT STREAM TEMPERATURE [C]	100.437266
OUTLET HOT STREAM PRESSURE [BAR]	14.311
INLET COLD STREAM TEMPERATURE [C]	81.5347563
INLET COLD STREAM PRESSURE [BAR]	15
OUTLET COLD STREAM TEMPERATURE [C]	91.5348052
OUTLET COLD STREAM PRESSURE [BAR]	14.655
OUTLET COLD STREAM VAPOR FRACTION	0.003320299
HEAT DUTY [WATT]	925993.5
CALCULATED HEAT DUTY [WATT]	925993.5
REQUIRED EXCHANGER AREA [SQM]	58.7104623
ACTUAL EXCHANGER AREA [SQM]	58.7104623
AVERAGE U (DIRTY) [WATT/SQM-K]	850
UA [J/SEC-K]	49903.8929
LMTD (CORRECTED) [C]	18.5555364
LMTD CORRECTION FACTOR	1
NUMBER OF SHELLS IN SERIES	1
TOTAL FEED STREAM CO2E FLOW [KG/HR]	12291.3175
TOTAL PRODUCT STREAM CO2E FLOW [KG/HR]	12291.3175

FLASH AND FLASH 2

NAME	FLASH	FLASH2
PROPERTY METHOD	ELECNRTL	ELECNRTL
HENRY'S COMPONENT LIST ID	GLOBAL	GLOBAL
ELECTROLYTE CHEMISTRY ID	GLOBAL	GLOBAL
USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO	NO
FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA	STEAM-TA
WATER SOLUBILITY METHOD	3	3
TEMPERATURE [C]		55
PRESSURE [BAR]	15	5
SPECIFIED VAPOR FRACTION		0.5
OUTLET TEMPERATURE [C]	81.5347563	55
OUTLET PRESSURE [BAR]	15	5
VAPOR FRACTION	0.002446836	0.855667194
HEAT DUTY [WATT]	0	-17003.3392
NET DUTY [WATT]		-17003.3392
FIRST LIQUID / TOTAL LIQUID	1	1
TOTAL FEED STREAM CO2E FLOW [KG/HR]	10789.7427	8971.96262
TOTAL PRODUCT STREAM CO2E FLOW [KG/HR]	10789.7427	8971.96262

PUMP, HP &LP

NAME	HP	LP	PUMP
PROPERTY METHOD	ELECNRTL	ELECNRTL	ELECNRTL
HENRY'S COMPONENT LIST ID	GLOBAL	GLOBAL	GLOBAL
ELECTROLYTE CHEMISTRY ID	GLOBAL	GLOBAL	GLOBAL
USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO	NO	NO
FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA	STEAM-TA	STEAM-TA
WATER SOLUBILITY METHOD	3	3	3
SPECIFIED DISCHARGE PRESSURE [BAR]	55	15	10
FLUID POWER [WATT]	170323.397	41090.6179	163.649984
CALCULATED BRAKE POWER [WATT]	238268.367	57433.1041	553.51178
ELECTRICITY [WATT]	238268.367	57433.1041	553.51178
VOLUMETRIC FLOW RATE [CUM/HR]	115.691364	116.477342	0.736424929
CALCULATED DISCHARGE PRESSURE [BAR]	55	15	10
CALCULATED PRESSURE CHANGE [BAR]	53	12.7	8
NPSH AVAILABLE [J/KG]	50.2283882	0.05307542	-197.635512
HEAD DEVELOPED [J/KG]	5356.30387	1293.23496	815.531055
PUMP EFFICIENCY USED	0.714838479	0.715451804	0.295657636
NET WORK REQUIRED [WATT]	238268.367	57433.1041	553.51178
TOTAL FEED STREAM CO2E FLOW [KG/HR]	1666.24798	1666.24798	13.2611049
TOTAL PRODUCT STREAM CO2E FLOW [KG/HR]	1666.24798	1666.24798	13.2611049

RADFRAC

NAME	ABSORBER	STRIPPER
PROPERTY METHOD	ELECNRTL	ELECNRTL
HENRY'S COMPONENT LIST ID	GLOBAL	GLOBAL
ELECTROLYTE CHEMISTRY ID	GLOBAL	GLOBAL
USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO	NO
FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA	STEAM-TA
WATER SOLUBILITY METHOD	3	3
NUMBER OF STAGES	15	18
CONDENSER	NONE	NONE
REBOILER	NONE	KETTLE
NUMBER OF PHASES	2	2
FREE-WATER	NO	NO
TOP STAGE PRESSURE [BAR]	53	2
CALCULATED MOLAR REFLUX RATIO	2.09419087	16.3850374
CALCULATED BOTTOMS RATE [KMOL/HR]	3838.32505	3637.18351
CALCULATED BOILUP RATE [KMOL/HR]	1990.51712	231.206535
CALCULATED DISTILLATE RATE [KMOL/HR]	1803.8839	231.731927
CONDENSER / TOP STAGE TEMPERATURE [C]	65.515192	75.8597586
CONDENSER / TOP STAGE PRESSURE [BAR]	53	2
CONDENSER / TOP STAGE REFLUX RATE [KMOL/HR]	3777.6772	3796.93628
REBOILER PRESSURE [BAR]	53	2.3
REBOILER TEMPERATURE [C]	82.0064138	109.304193
REBOILER HEAT DUTY [WATT]		6888022.08
TOTAL FEED STREAM CO2E FLOW [KG/HR]	14479.1128	10638.3307
TOTAL PRODUCT STREAM CO2E FLOW [KG/HR]	14479.1128	10638.2106
NET STREAM CO2E PRODUCTION [KG/HR]		-0.120071989
TOTAL CO2E PRODUCTION [KG/HR]		-0.120071989
CALCULATED MOLAR BOILUP RATIO		0.063567465
CALCULATED MASS BOILUP RATIO	0.221342188	0.061336799

MIXER

NAME	MIXER1	MIXER2
PROPERTY METHOD	ELECNRTL	ELECNRTL
HENRY'S COMPONENT LIST ID	GLOBAL	GLOBAL
ELECTROLYTE CHEMISTRY ID	GLOBAL	GLOBAL
USE TRUE SPECIES APPROACH FOR ELECTROLYTES	NO	NO
FREE-WATER PHASE PROPERTIES METHOD	STEAM-TA	STEAM-TA
WATER SOLUBILITY METHOD	3	3
OUTLET TEMPERATURE [C]	51.6131134	100.508531
CALCULATED OUTLET PRESSURE [BAR]	2	2
VAPOR FRACTION	0.000750606	0
FIRST LIQUID /TOTAL LIQUID	1	1
TOTAL FEED STREAM CO2E FLOW [KG/HR]	13.2611281	1666.24798
TOTAL PRODUCT STREAM CO2E FLOW [KG/HR]	13.2611049	1666.24798
NET STREAM CO2E PRODUCTION [KG/HR]	-2.32E-05	0

Problem Analysis

In order to achieve 1 mol% of CO_2 in the clean gas stream, the absorber requires 3642.209 kmol/hr of lean solvent and the stripper requires a reboiler duty of 6888022.08 Watts.

	Duty(Watts)	Duty(GJ/hr) or (kWh/hr)	Utilities Cost ($/GJ)	Total($/hr)
REBOILER	6888022	24.7969	5.5	$ 136.38
CONDENSER	-434088	-1.5627	0.2	$ 0.31
COOLER	-6142853	-22.114	0.2	$ 4.42
heatx	925993.5	3.33358	5.5	$ 18.33
PUMP	553.512	0.55351	0.06	$ 0.03
LP	57433.104	57.4331	0.06	$ 3.45
HP	238268.367	238.268	0.06	$ 14.30
Flash2	-17003.3	-0.0612	0.2	$ 0.01

Sum($/hr) = $177.24 per hr

*1 Year = 8000 operating hours

Annual Operating Expenses: $ 1,417,923.57

From the calculations above, the total annual operating cost for this process is $1,417,923.57